

MoonAP v0.1: A MoonBit Agent Playground for LLM-Generated Browser-Local WebAssembly Applications

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Abstract

Large language models (LLMs) can synthesize useful programs from natural-language descriptions, but the path from generated code to a runnable, inspectable, and reusable application remains fragmented. This paper presents MoonAP v0.1, a MoonBit-oriented agent playground that turns a natural-language application request into MoonBit source code, compiles it to WebAssembly, executes it locally in the browser, and optionally saves the successful artifact as a reusable MoonAP-SKILL. The system combines an LLM routing layer, a MoonBit compilation service, browser-local runtime contracts, and a three-layer SKILL architecture for personal reuse, local installation, and cloud discovery. The v0.1 prototype demonstrates twelve task categories, including form-based calculators, JSON and text utilities, CSV file analysis, and browser-local large FastQ generation and analysis. By moving file handling and computation into the browser, MoonAP keeps raw user data on the client, exploits edge-side compute resources, and reduces repeated cloud LLM token consumption after a useful artifact has been created. We describe the system design, implementation, runtime contract abstraction, and lessons learned from building an end-to-end software synthesis workflow around a young programming language and browser-local execution. The central finding is not that current lightweight LLMs can already synthesize MoonBit reliably; rather, it is that a practical synthesis environment needs explicit runtime contracts, deterministic simulation paths, artifact reuse, and local execution boundaries to make generated applications usable.

1 Introduction

LLM-based software synthesis has made it possible for users to describe a program in natural language and receive executable source code [2, 1, 3]. However, code generation alone is rarely a complete user workflow. A practical system must select or configure a model, preserve prompts and generated source, compile the program, expose runtime inputs, execute the artifact, present results, and allow successful programs to be reused. These requirements become more pronounced when the target language is new, when the generated program is intended to run in a browser, or when user data should remain local.

MoonAP is a MoonBit Agent Playground designed to explore this complete workflow. A user describes an application, the system asks an LLM to generate MoonBit code, a local server compiles the code to WebAssembly, and the browser runs the resulting application with a task-specific runtime UI. If the result is useful, the application can be saved as a MoonAP-SKILL and reused later without another LLM call.

MoonBit is a young programming language with a toolchain oriented toward modern application targets, including WebAssembly [4, 6]. This makes it attractive for browser-local synthesized

applications, but it also exposes a difficulty: many lightweight or free LLM routes do not yet generate valid MoonBit code consistently. MoonAP therefore treats code generation as only one part of the system. It pairs real API routes with deterministic simulated routes, embeds runtime specifications in generated artifacts, and separates generation-time behavior from reuse-time behavior.

This paper reports on MoonAP v0.1 as a system prototype. The contribution is a working architecture for connecting LLM-generated MoonBit code to browser-local execution and reuse:

- an end-to-end pipeline from natural-language task to MoonBit source, WebAssembly compilation, browser-local execution, and reusable SKILL;
- a runtime contract abstraction for generated applications, covering form, text, report, file, and large-file workflows;
- a three-layer SKILL model that distinguishes personal reuse, locally installed public SKILLS, and cloud discovery;
- browser-local large-file workflows for FastQ generation and analysis, including streamed and chunked processing paths designed for GB-scale files whose raw contents remain on the client;
- an implementation and qualitative evaluation over twelve v0.1 demonstration categories.

2 Motivation and Design Goals

MoonAP began from a concrete need: a user should be able to ask for a small application, run it immediately, and keep it if it works. The system was also motivated by a privacy-sensitive large-file case study: generating and analyzing FastQ sequencing-like files in the browser without uploading file contents to an LLM. Over time, the project evolved from a domain-specific demonstration into a general agent playground.

The design goals of v0.1 are as follows.

MoonBit-first synthesis. The target language of generated applications is MoonBit. The system should keep MoonBit source, compiler output, WebAssembly artifacts, and runtime metadata visible enough to support debugging and reuse.

Browser-local execution. Generated applications should run in the user’s browser where possible. This is especially important for file-oriented workflows, where local files should not be sent to the LLM.

Reusable artifacts. A successful run should not be a one-off event. The generated application should be saved as a structured package that can be reopened and executed later.

Separation of generation and runtime. The LLM can generate source code and runtime metadata, but the browser host should control file pickers, local storage, large-file streaming, and presentation of results.

Stable demonstration path. Because real LLM providers vary in availability and MoonBit capability, the system should include a deterministic simulated route for repeatable demonstrations and regression checks.

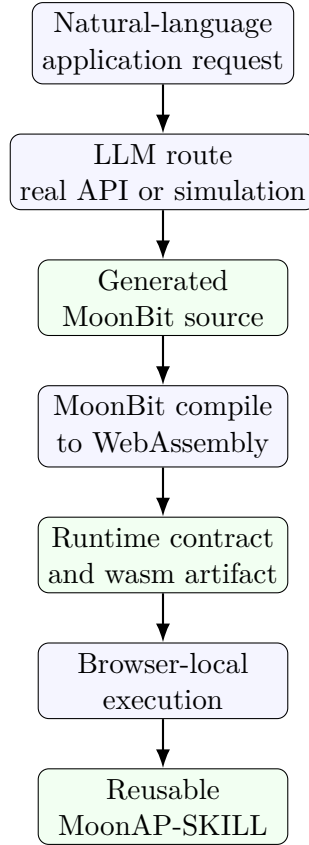


Figure 1: End-to-end MoonAP workflow. The LLM participates in source generation, while browser-local execution and artifact reuse are controlled by the host system.

3 System Overview

Figure 1 shows the MoonAP pipeline. A browser UI accepts a natural-language task. The local server routes the request to a real LLM provider or a simulated LLM path. The resulting MoonBit source is compiled to WebAssembly using the MoonBit toolchain. The browser then renders a runtime UI from the artifact’s runtime contract, executes the browser-local step, and records the result. If the user accepts the result, the app is materialized as a reusable SKILL.

4 MoonAP-SKILL Architecture

MoonAP-SKILLs are reusable application packages. A saved SKILL contains a human-readable `SKILL.md`, generated MoonBit source, a compiled WebAssembly artifact, runtime metadata, and optional distribution files. The v0.1 design separates SKILLs into three layers (Figure 3).

This separation addresses two practical issues. First, it prevents cloud catalog entries from being treated as directly executable objects before they have been installed and inspected locally. Second, it gives the user a clear distinction between self-created artifacts and public artifacts obtained from a hub. Figure 4 shows the browser UI for selecting a Local-SKILL-Hub directory, inspecting installed SKILLs, refreshing the Cloud-SKILL-Hub catalog, installing public SKILLs, and running local SKILLs.

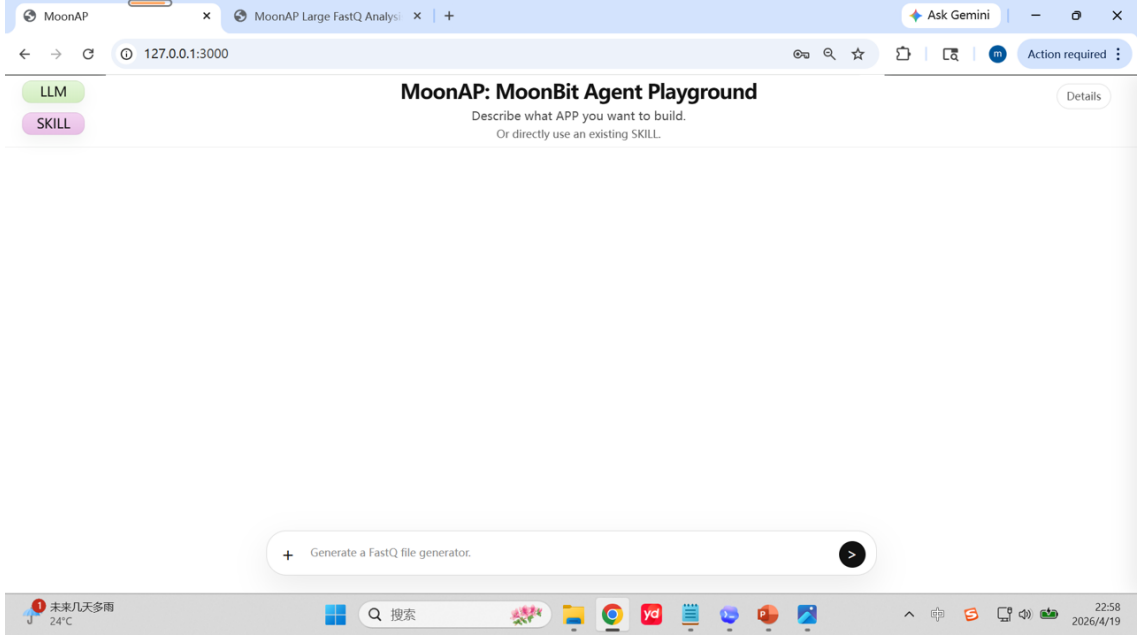


Figure 2: MoonAP browser UI. Users can enter a natural-language application request or switch to the SKILL interface for reuse and installation.

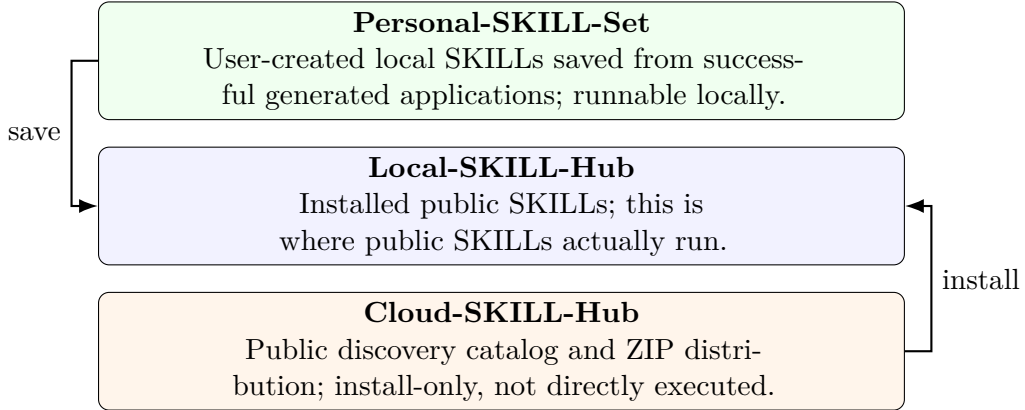


Figure 3: Three-layer MoonAP-SKILL model. Cloud entries are used for discovery and installation; execution happens in Personal or Local stores.

5 Runtime Contracts

One of the central lessons from v0.1 is that synthesized source code is not enough to produce a usable application. The host must know what input controls to render, whether an input file is required, how results should be displayed, and whether the output should be downloaded, shown as a report, or treated as an interactive view.

MoonAP uses runtime contracts for this purpose. A runtime contract is a structured specification associated with a generated artifact. It can include:

- runtime mode, such as `form`, `file`, or `interactive`;
- result mode, such as `text`, `report`, or `download`;
- input fields with labels, types, default values, and validation hints;
- an I/O contract for file or large-file workflows;

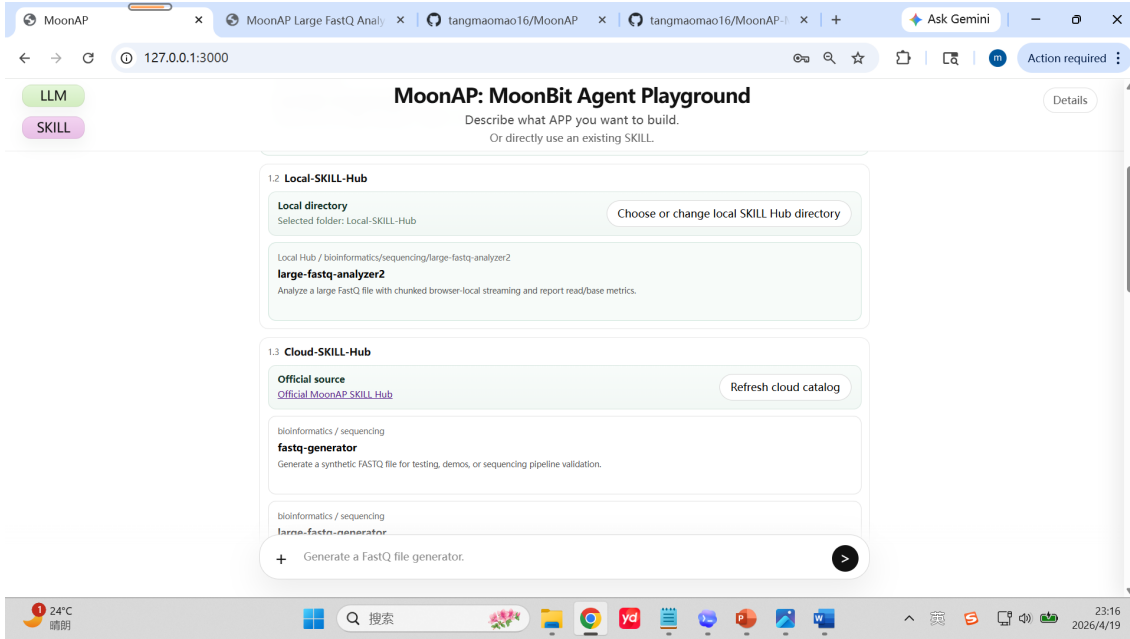


Figure 4: MoonAP Local-SKILL-Hub and Cloud-SKILL-Hub interface. Installed public SKILLs run from the local hub, while cloud entries are used for discovery and installation.

- action labels and rerun labels for the UI;
- tool-specific metadata, such as CSV summary or FastQ analysis.

The runtime contract enables generic execution paths for many small tools. For example, the Celsius/Fahrenheit converter, minutes/seconds converter, BMI calculator, circle calculator, loan payment calculator, tip calculator, JSON formatter, and text analyzer can all be represented as form or text contracts. The CSV analyzer uses a file contract with a browser-local file picker and report output.

The contract also makes SKILL reuse possible. When a SKILL is run again, MoonAP does not need to ask the LLM to regenerate code. Instead, it reloads the saved runtime specification, renders the appropriate UI, and executes the local runtime path.

6 Browser-Local Large-File Workflows

MoonAP v0.1 includes two large-file FastQ workflows: generation and analysis. These workflows are intentionally treated as high-value demonstrations of browser-local computation rather than as the only purpose of the platform. They also illustrate an important design point for LLM-generated applications: the LLM should help create the tool, but it should not need to ingest the user’s raw data file.

Large FastQ generation. The generator uses a browser file-save picker and a streamed writer. The user chooses an output file, enters parameters such as target size, read length, random seed, and N-base rate, and the browser writes records locally. This design can produce large synthetic files without constructing the whole file in server memory, and the LLM does not receive the generated file contents.

Large FastQ analysis. The analyzer uses a browser file-open picker and reads the selected file in chunks. It counts reads, lines, bases, base composition, malformed records, read lengths, and

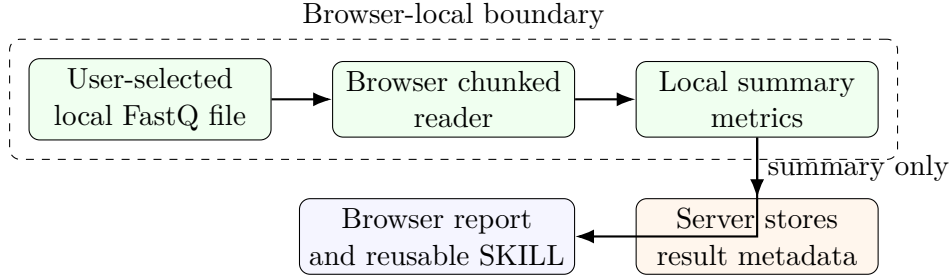


Figure 5: Browser-local large-file analysis. File contents are processed in the browser; only summary metadata is recorded by the local server.

The screenshot shows a web browser window with a 'FastQ Summary' report. The report includes a table of summary metrics and the beginning of a 'Base Composition' table.

FastQ Summary	
Estimated read count	3307239
Total lines	13228956
Total bases	496085850
Malformed records	0
Min read length	150
Max read length	150
Average read length	150
Elapsed ms	20736

Base Composition		
Base	Count	Percent
A	133706384	24.75%

Figure 6: FastQ analysis report with summary metrics and the beginning of the base-composition table. The counts are computed locally in the browser from the selected FastQ file.

elapsed time. Because the raw bytes are processed through browser-local chunking, the workflow is suitable for large local files, including GB-scale demonstrations on commodity machines. File contents stay in the browser; the server receives only summary metrics and result metadata.

Figure 6 shows the browser-local FastQ report generated by the analyzer. The report is opened from a browser blob URL and presents summary metrics and base-composition results without uploading the original file contents.

7 Implementation

The prototype is organized as a MoonBit project with a local server package and a browser application package. The server handles HTTP endpoints, LLM routing, runtime registration, artifact persistence, SKILL export, and static file serving. The browser app implements the user interface, runtime cards, file pickers, generic runtime execution, large-file progress reporting, SKILL installation, and SKILL reuse.

The system currently targets a Windows-first release path. Native server builds use the MoonBit toolchain together with MSVC setup scripts. The browser side is compiled to JavaScript and committed as distribution bundles for fresh-clone startup fallback. The primary validated

Capability family	Examples	Runtime path
Computed form apps	Celsius/Fahrenheit, minutes/seconds, two-number sum, BMI, circle, loan, tip	Generic form contract
Text and JSON utilities	JSON formatter/validator, text analyzer	Generic text runtime
Browser-local file utility	CSV summary analyzer	Generic file/report runtime
Large-file generation	Large FastQ generator	Streamed browser writer
Large-file analysis	Large FastQ analyzer	Chunked browser reader/report
Reuse	Personal SKILL, Local-SKILL-Hub	Saved runtime spec
Sharing/install	Cloud-SKILL-Hub	Catalog and ZIP install

Table 1: MoonAP v0.1 demonstration matrix.

browser target is Chrome or Edge, because large-file workflows depend on the File System Access API [7].

7.1 LLM Routes

MoonAP separates real and simulated LLM routes. Real routes support multiple providers or OpenAI-compatible endpoints. The simulated route provides deterministic outputs for the v0.1 demo set. This distinction is important because real lightweight models often fail to produce valid MoonBit in the current ecosystem. The simulated route is not presented as an evaluation of model quality; it is a system-testing path that exercises the compile, runtime, and SKILL reuse pipeline.

7.2 Artifact Lifecycle

The lifecycle of an artifact has four stages:

1. **Generation:** prompt and LLM output are captured.
2. **Compilation:** MoonBit source is compiled into a WebAssembly artifact.
3. **Runtime:** the browser executes a contract-driven runtime step and records a result.
4. **Reuse:** the artifact is materialized as a SKILL and later run from Personal-SKILL-Set or Local-SKILL-Hub.

This lifecycle allows MoonAP to treat a successful generated app as an asset rather than a transient response.

8 Evaluation and Case Studies

The v0.1 evaluation is qualitative and prototype-oriented. It asks whether the system can express, run, and reuse representative browser-local applications. Table 1 summarizes the current demonstration matrix.

The small utility tasks exercise the generic contract path. They show that the browser can render task-specific controls without bespoke UI code for each generated app. The CSV analyzer extends this pattern to file input and report output. The FastQ generator and analyzer exercise specialized large-file runtime paths that rely on browser-local streaming and chunking.

The most important engineering result is the reuse path. A successful generated app can be saved as a SKILL, reopened, and executed from a local store without another LLM call. This substantially changes the user experience: the LLM is used to create or repair an artifact, but the artifact can later behave like a normal local application.

9 Lessons Learned

Generated code needs a host contract. Without a runtime contract, the browser does not know which inputs to show or how to interpret results. Embedding or saving this contract makes generated apps reusable.

A deterministic path is necessary for system development. When the target language is young, real models may fail for reasons unrelated to the runtime system. The simulated route allowed v0.1 to test compilation, UI, file handling, and SKILL reuse deterministically.

Reuse is as important as generation. Users should not need to pay the latency and uncertainty cost of an LLM call every time they run a useful generated app. The SKILL format turns synthesis into a reusable software artifact. This also reduces cloud token consumption: the expensive generative step is used for creation or repair, while repeated execution happens locally from the saved contract and compiled artifact.

Browser-local file boundaries must be explicit. For large files, privacy and performance depend on a clear boundary: file contents stay in the browser, while summaries and metadata can be recorded by the local server.

Client hardware is part of the execution platform. Browser-local WebAssembly execution lets MoonAP use computation and storage resources already available on the user’s device. This edge-side execution model is especially valuable for file utilities, where moving raw data to a cloud model would be slower, more expensive, and less private than processing it near the data.

10 Limitations

MoonAP v0.1 has several limitations. First, real LLM generation of valid MoonBit is not yet reliable across lightweight or free models. Stronger models may perform better, but model capability and provider availability are external variables. Second, the current release path is best tested on Windows with Chrome or Edge. Third, the generic runtime contract is still small and does not yet cover arbitrary application UIs, complex stateful apps, or a universal WebAssembly host ABI. Fourth, the evaluation is a prototype case study rather than a large-scale benchmark. Finally, security hardening remains future work: generated code, saved artifacts, and hub-installed packages need stronger provenance and sandboxing mechanisms before broad deployment, a concern consistent with prior work on the security risks of generated code [5].

11 Future Work

Future versions of MoonAP can extend the platform in several directions:

- a more formal runtime contract schema and validation layer;
- a stable host ABI for generated WebAssembly programs with file, stream, and UI callbacks;
- stronger automated repair loops for MoonBit compilation errors;

- cross-platform server support beyond the current Windows-first path;
- richer SKILL provenance, signatures, versioning, and dependency management;
- quantitative evaluation over larger prompt suites and real model providers;
- improved UI generation for interactive applications and games.

12 Conclusion

MoonAP v0.1 demonstrates an end-to-end MoonBit software synthesis workflow: natural-language task descriptions become MoonBit source, compiled WebAssembly artifacts, browser-local applications, and reusable SKILL packages. The system highlights a practical lesson for LLM-based software synthesis: generation is only the first step. A usable platform also needs runtime contracts, local execution boundaries, artifact packaging, deterministic testing paths, and reuse mechanisms. By keeping raw data on the client, using browser-side computation for file workflows, and reusing saved SKILLs without repeated generation, MoonAP points toward a lower-token and more privacy-preserving style of practical software synthesis. Although current lightweight LLMs do not yet generate MoonBit reliably, MoonAP shows how a MoonBit-centered agent playground can make generated applications inspectable, runnable, and reusable in the browser.

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